IGCSE 0607 Practice Questions — E1.2 — Combined Questions & Markscheme

Sets

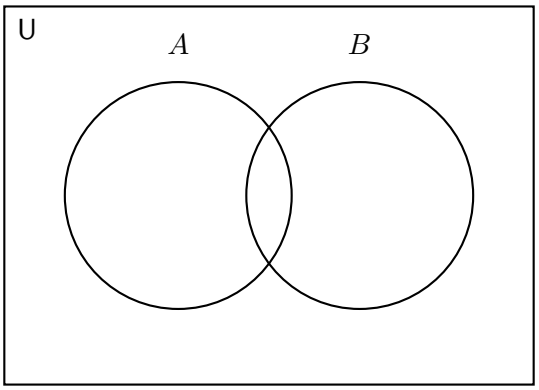
Total Questions: 18 — **Total Marks:** 112 Marks

Table of Contents & Quick Navigation

- **E1.2** Sets
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

1 The universal set U contains all integers from 1 to 10 inclusive.
Set A contains all the even numbers in U.
Set B contains all the prime numbers in U.

(a) Complete the Venn diagram to show the elements of these sets.



[2]

(b) List the elements of $A \cap B$.

..... **[1]**

(c) Find $n(A \cup B)'$.

..... **[1]**

2 $\mathsf{U} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$
$A = \{x : x \text{ is a prime number}\}$
$B = \{x : x \text{ is a factor of } 12\}$

(a) List the members of set A.

..... **[1]**

(b) List the members of set B.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

(d) Write down the elements of $A \cap B$.

..... **[1]**

(e) Write down a number that is in U but is not in $A \cup B$.

..... **[1]**

3 The universal set U and sets A and B are defined as follows:

- $\mathsf{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- $A = \{2, 3, 5, 7\}$
- $B = \{1, 3, 5, 7, 9\}$

(a) List the elements of $A \cap B$.

..... **[1]**

(b) Find $n((A \cup B)')$.

..... **[2]**

(c) Write down the correct set notation symbol $(\in, \notin, \subset, \cup, \cap)$ to complete the statement:

$2 \ldots A$

[1]

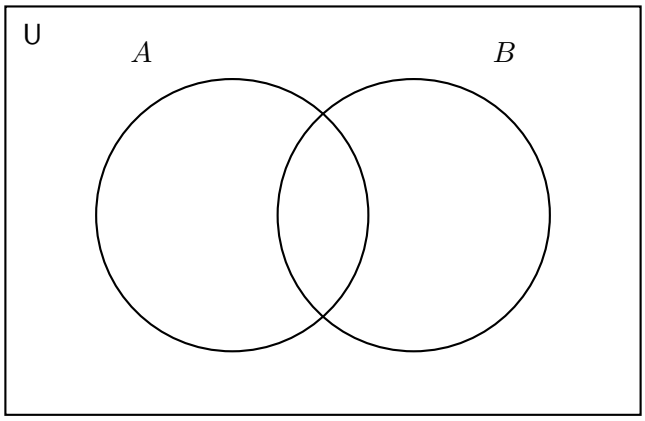
(d) Write down the correct set notation symbol to complete the statement:

$\{1, 9\} \ldots B$

[1]

4 Let U be the set of integers from 1 to 10 inclusive.
Let A be the set of prime numbers.
Let B be the set of odd numbers.

(a) Complete the Venn diagram to show the elements of these sets.



[2]

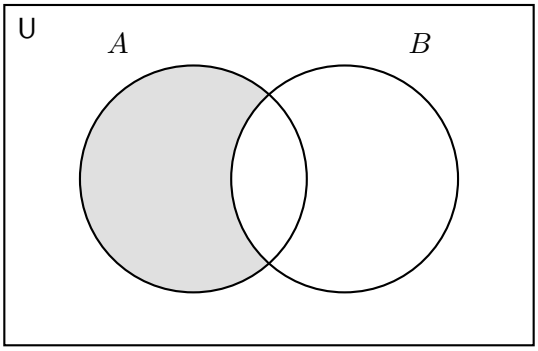
(b) List the elements of $A \cap B'$.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

(d) Write down the set notation for the shaded region in the Venn diagram above.



..... **[1]**

5 The universal set U and sets P and F are defined as follows:

- $\mathsf{U} = \{x : x \text{ is an integer} 1 \leq x \leq 10\}$
- $P = \{x : x \text{ is a prime number}\}$
- $F = \{x : x \text{ is a factor of } 18\}$

(a) Write down the elements of set F.

$F = \{$ ..... $\}$ **[1]**

(b) List the elements of $P \cap F$.

$P \cap F = \{$ ..... $\}$ **[1]**

(c) Find $n((P \cup F)')$.

..... **[2]**

(d) Write down the correct set symbol, $\in$ or $\notin$, to make this statement true:

$5 \ldots\ldots F$

[1]

6 The universal set U and sets A and B are defined as follows:

$\mathsf{U} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$

$A = \{x : x \text{ is a prime number}\}$

$B = \{x : x \text{ is a factor of } 10\}$

(a) List the elements of set A.

..... **[1]**

(b) Find $n(A \cup B)$.

..... **[2]**

(c) Complete each statement by inserting the correct symbol from $\in, \notin, \subseteq, \cup, \cap$.

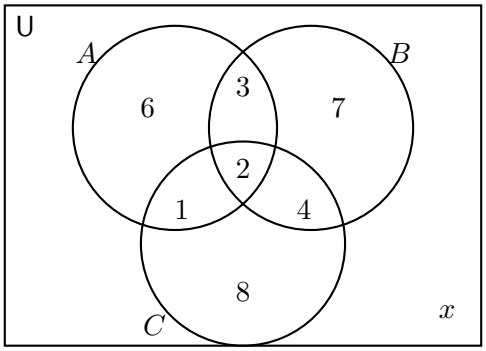
(i) 4 ..... A

[1]

(ii) $A \cap B$ ..... A

[1]

7 The Venn diagram shows the number of elements in each region for sets A, B, and C.



(a) Given that $n(\mathsf{U}) = 35$, find the value of x.

$x =$ ..... **[2]**

(b) Find $n(A \cap C')$.

..... **[1]**

(c) Use set notation to describe the set of elements whose number is 4.

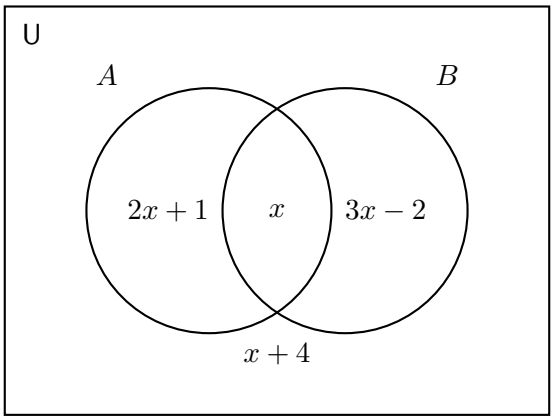
..... **[1]**

(d) Find $n(A \cup (B \cap C))$.

..... **[2]**

8 The Venn diagram shows the number of elements in sets A and B, in terms of x.

U is the universal set and $n(\mathsf{U}) = 52$.



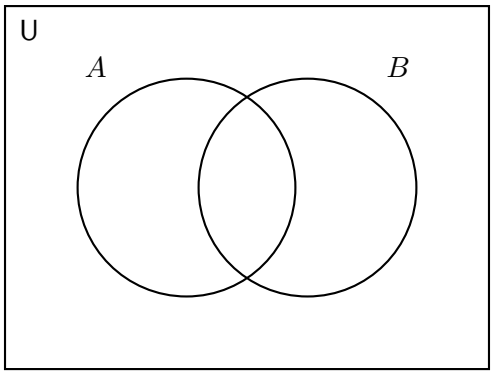
(a) Form an equation in x and solve it.

$x =$ ..... **[3]**

(b) Find $n(A \cup B')$.

..... **[2]**

(c) On the Venn diagram below, shade the region $A \cap B'$.



[1]

9 The universal set U is the set of all real numbers.
Four sets P, Q, R and T are defined as follows:

$P = \{x : -2 < x \leq 5\}$

$Q = \{x : -4 \leq x < 3\}$

$R = \{x : x > 1\}$

$T = \{x : x \leq -1 \text{ or } x \geq 2\}$

(a) Write down the set $P \cap Q$ as an inequality.

..... **[2]**

(b) Find the integer elements of the set $P \cap R'$.

..... **[2]**

(c) Let $S = \{x : x \leq k\}$.
Given that $P \cap S = P$, find the range of possible values for k.

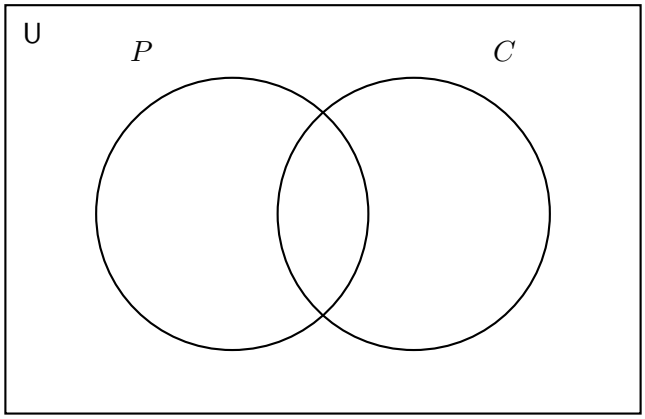
..... **[2]**

(d) Write down the set $Q \cap T$ as inequalities.

..... **[2]**

10 There are 45 students in a class.
Let U be the set of students in this class.
Let P be the set of students who study Physics.
Let C be the set of students who study Chemistry.
$n(P) = 28$
$n(C) = 22$
$n(P \cap C) = x$
$n((P \cup C)') = 4$

(a) Complete the Venn diagram in terms of x.



[2]

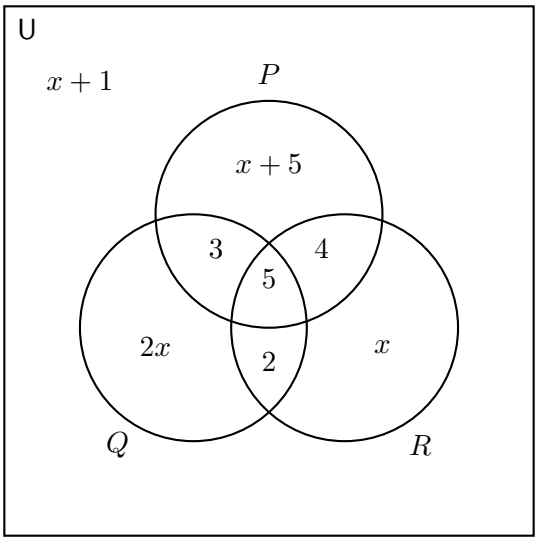
(b) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[2]**

(c) Find $n(P \cap C')$.

..... **[1]**

11 The Venn diagram shows the number of elements in each region for the sets P, Q and R.



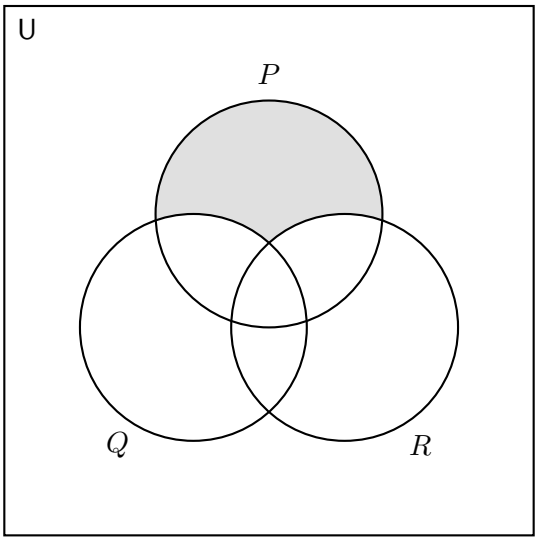
(a) The total number of elements in the universal set is 45.
Form an equation in terms of x and solve it to find the value of x.

$x =$ ..... **[3]**

(b) Find $n(P \cup Q)$.

..... **[1]**

(c) On the Venn diagram below, shade the region $P \cap (Q \cup R)'$.

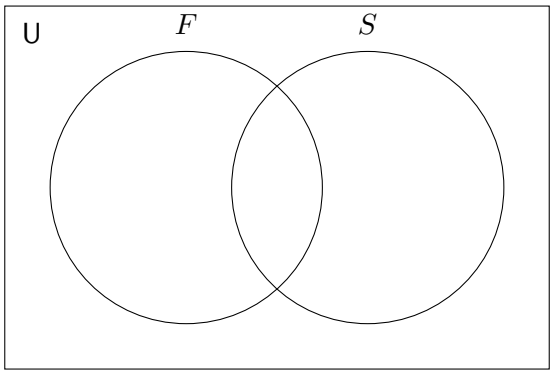


[1]

12 In a class of 40 students:

- 25 study French (F)
- 18 study Spanish (S)
- 5 study neither French nor Spanish

(a) Complete the Venn diagram to show this information.



[3]

(b) Find $n(F \cap S')$.

..... **[1]**

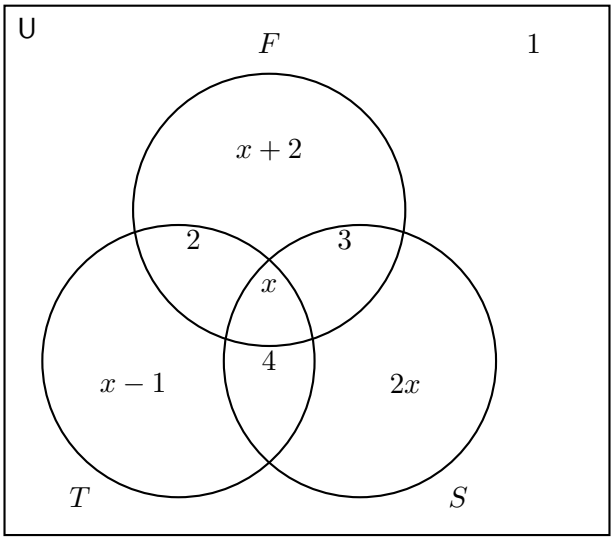
(c) A student is chosen at random from the class.
Find the probability that they study exactly one of the two languages.

..... **[2]**

(d) Two students are chosen at random from those who study French.
Find the probability that both also study Spanish.

..... **[1]**

13 A group of 36 students were asked which sports they play.
The sports are Football (F), Tennis (T) and Swimming (S).
The Venn diagram shows the number of students in each region, in terms of x.



(a) Write down an equation in x and solve it to find the value of x.

$x =$ ..... **[3]**

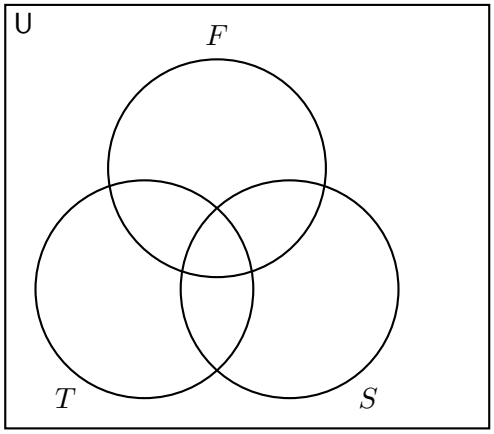
(b) Find the number of students who play Football.

..... **[2]**

(c) Find $n(T' \cap S)$.

..... **[2]**

(d) On the Venn diagram, shade the region $F \cap (T \cup S)'$.



[1]

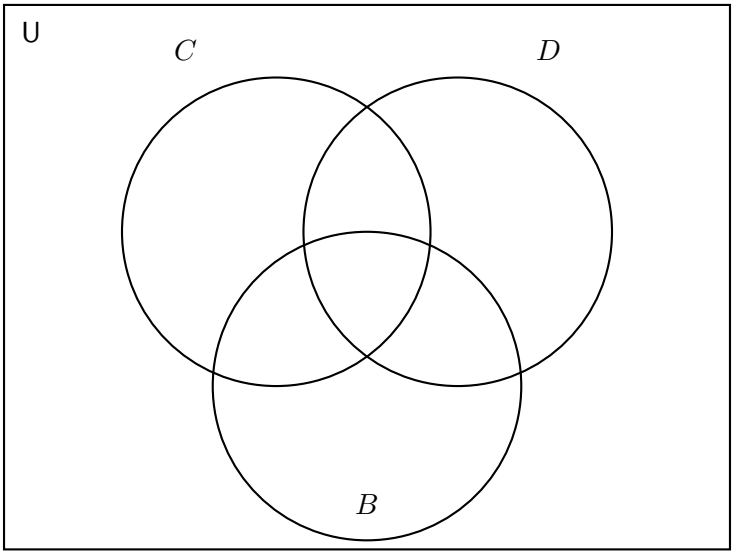
14 In a survey of 100 people, information was collected about the pets they own.

Let U be the set of people surveyed.
Let C be the set of people who own a cat.
Let D be the set of people who own a dog.
Let B be the set of people who own a bird.

- 8 people own all three pets.
- 12 people own a cat and a dog but not a bird.
- 7 people own a cat and a bird but not a dog.
- 10 people own a dog and a bird but not a cat.
- 3 people do not own any of these pets.

The number of people who own only a cat is x.
The number of people who own only a bird is also x.
The number of people who own only a dog is $2x$.

(a) Complete the Venn diagram to show this information.



[3]

(b) Find the value of x.

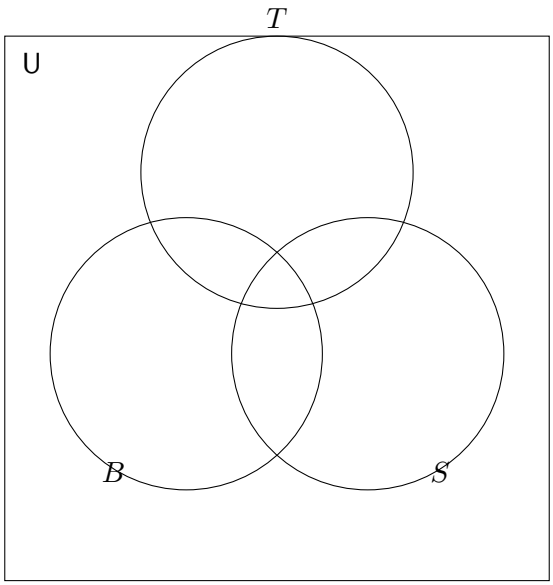
$x =$ ..... **[2]**

(c) Find $n((C \cup D) \cap B')$.

..... **[2]**

15 There are 60 people in a sports club.

- 30 play Tennis (T)
- 28 play Badminton (B)
- 25 play Squash (S)
- 12 play both Tennis and Badminton
- 9 play both Tennis and Squash
- 11 play both Badminton and Squash
- 4 play all three sports



(a) Complete the Venn diagram.

[3]

(b) Find $n(T \cup B \cup S)'$.

..... **[1]**

(c) Find $n(T \cap (B \cup S))$.

..... **[1]**

(d) Use set notation to describe the region containing the number 9 (the number of people who play Badminton only).

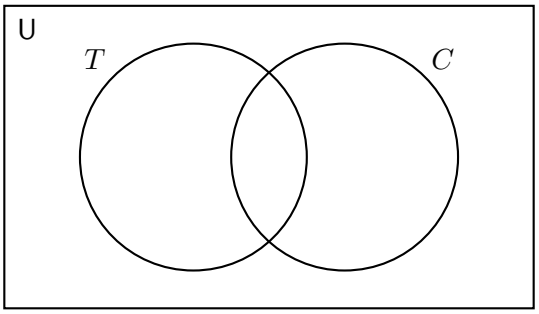
..... **[2]**

16 In a sports club, the universal set U represents all the students.
The set T represents students who play Tennis.
The set C represents students who play Cricket.

You are given the following information:

- $n(\mathsf{U}) = 50$
- $n(T) = 3x$
- $n(C) = 4x$
- $n(T \cap C) = x$
- $n((T \cup C)') = 2$

(a) Complete the Venn diagram, writing the number of students in each region in terms of x where appropriate.



[3]

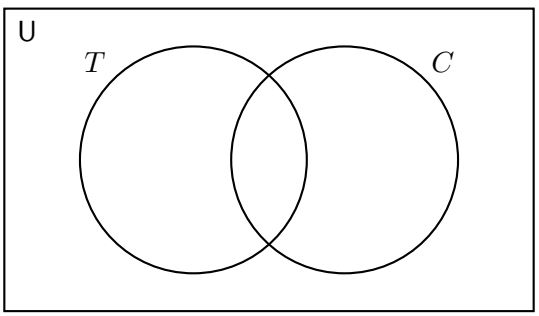
(b) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[2]**

(c) Find the number of students who play Tennis but not Cricket.

..... **[1]**

(d) On the Venn diagram below, shade the region $T' \cup C$.

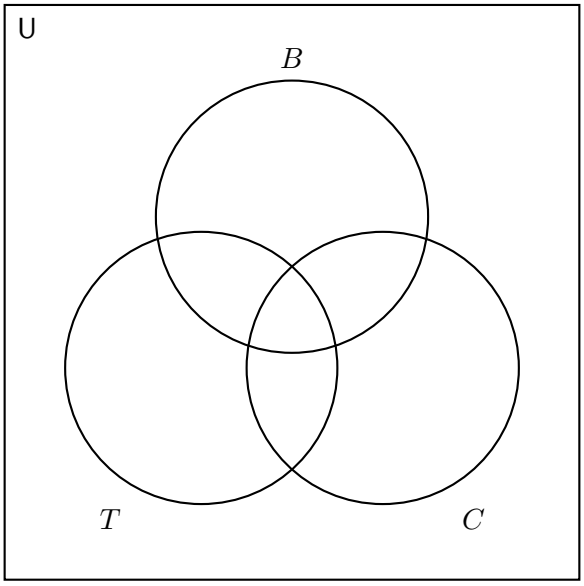


[2]

17 In a survey of 103 people, the following information was recorded about their use of three types of transport: Bus (B), Train (T), and Car (C).

- 15 people use all three types.
- 28 people use both a Train and a Car.
- 22 people use both a Bus and a Car.
- 30 people use both a Bus and a Train.
- x people use only a Bus.
- The number of people who use only a Train is twice the number of people who use only a Bus.
- The number of people who use only a Car is 5 more than the number of people who use only a Bus.
- 8 people do not use any of these types of transport.

(a) Complete the Venn diagram in terms of x to show this information.



[3]

(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find $n(T \cup C)$.

..... **[1]**

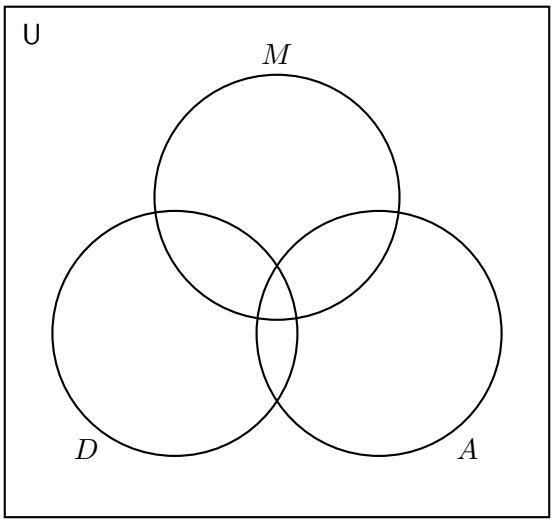
(d) A person is chosen at random from those who use a Train.
Find the probability that this person also uses a Car but does not use a Bus.

..... **[2]**

18 In a year group of 120 students, they can choose to study Music (M), Drama (D), and Art (A).

- 12 students study all three subjects.
- 25 students study Music and Drama.
- 32 students study Drama and Art.
- 28 students study Music and Art.
- 50 students study Music.
- 65 students study Drama.
- x students study Art only.
- 15 students study none of these three subjects.

(a) Use this information to complete the Venn diagram.



[3]

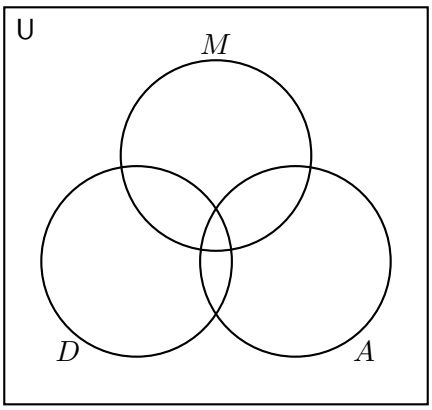
(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find $n(M \cup (D \cap A))$.

..... **[2]**

(d) On the Venn diagram below, shade the region $M \cap (D \cup A)'$.



[1]

1 **(a)** Correct elements in all four regions: 2 in the intersection, $4, 6, 8, 10$ in A only, $3, 5, 7$ in B only, and $1, 9$ outside both circles **[B2]**

If 0 scored, allow [B1] for 2 or 3 correct regions

(b) $\{2\}$ or 2 **[B1]**

(c) 2 **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** 2, 3, 5, 7 **[B1]**

(b) 1, 2, 3, 4, 6, 12 **[B1]**

(c) 8 **[B1]**

(d) 2, 3 **[B1]**

(e) 8 or 9 or 10 **[B1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $\{3, 5, 7\}$ **[B1]**

(b) 4 **[B2]**

M1 for correctly identifying $(A \cup B)' = \{4, 6, 8, 10\}$ or $A \cup B = \{1, 2, 3, 5, 7, 9\}$.

(c) $\in$ **[B1]**

(d) $\subset$ or $\subseteq$ **[B1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** Intersection: $3, 5, 7$ **[B1]**

Remainder of diagram correct: A only region has 2; B only region has $1, 9$; Outer region has $4, 6, 8, 10$ **[B1]**

(b) 2 **[B1]**

(c) 6 **[B1]**

(d) $A \cap B'$ **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $1, 2, 3, 6, 9$ **[B1]**

(b) $2, 3$ **[B1]**

(c) 3 **[B2]**

Partial working: Correctly identifies elements in the union $\{1, 2, 3, 5, 6, 7, 9\}$ OR elements outside $\{4, 8, 10\}$ **[M1]**

(d) $\notin$ **[B1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $2, 3, 5, 7$ **[B1]**

(b) Listing elements of $A \cup B$ as $1, 2, 3, 5, 7, 10$ **[M1]**

6 **[A1]**

(c) **(i)** $\notin$ **[B1]**

(ii) $\subseteq$ **[B1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** 4 **[B2]**

M1 for $6 + 3 + 7 + 1 + 2 + 4 + 8 + x = 35$ or $31 + x = 35$.

(b) 9 **[B1]**

(c) $A' \cap B \cap C$ or $B \cap C \cap A'$ **[B1]**

(d) 16 **[B2]**

M1 for correctly identifying $n(B \cap C) = 6$ or summing $12 + 4$.

[Back to Question] *[Total Marks: 6]*

8 **(a)** 7 **[B3]**

Partial working: Correct simplified equation $7x + 3 = 52$ **[M2]**

Partial working: Sets up unsimplified sum $(2x + 1) + x + (3x - 2) + (x + 4) = 52$ **[M1]**

(b) 33 **[B2]**

Partial working: Evaluates correct sub-regions for $x = 7$ (e.g. $15 + 7 + 11$) OR identifies correct algebraic expression $4x + 5$ **[M1]**

(c) Region strictly inside A but outside B is shaded (the left crescent) **[B1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** Identifies correct boundary values -2 and 3 **[M1]**

$-2 < x < 3$ **[A1]**

(b) Identifying $R' = \{x : x \leq 1\}$ leading to intersection $-2 < x \leq 1$ **[M1]**

$-1, 0, 1$ **[A1]**

(c) Recognising that $P \cap S = P$ implies $P \subseteq S$ **[M1]**

$k \geq 5$ **[A1]**

(d) Identifies one correct distinct interval, e.g., $-4 \leq x \leq -1$ or $2 \leq x < 3$ **[M1]**

$-4 \leq x \leq -1$ and $2 \leq x < 3$ (accept 'or' between them) **[A1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** Intersection: x and Outer region: 4 **[B1]**

P only region: $28 - x$ AND C only region: $22 - x$ **[B1]**

(b) $(28 - x) + x + (22 - x) + 4 = 45$ oe **[M1]**

$x = 9$ **[A1]**

(c) 19 **[B1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** Summing all regions: $(x + 5) + 2x + x + 3 + 4 + 2 + 5 + (x + 1) = 45$ (or equivalent) **[M1]**

$5x + 20 = 45 \implies 5x = 25$ **[M1]**

$x = 5$ **[A1]**

(b) 34 **[B1]**

(c) Shading exactly the region representing P only (the top part of circle P, excluding the intersections with Q and R) **[B1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** 5 in outer region **[B1]**

8 in intersection **[B1]**

17 in F only AND 10 in S only **[B1]**

(b) 17 **[B1]**

(c) $\frac{27}{40}$ **[B2]**

M1 for $17 + 10$ or 27 seen **[M1]**

(d) $\frac{8}{25} \times \frac{7}{24}$ **[M1]**

$\frac{7}{75}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

13 **(a)** $(x + 2) + 2 + 3 + (x - 1) + 4 + 2x + x + 1 = 36$ **[M1]**

$5x + 11 = 36$ or $5x = 25$ **[M1]**

5 **[A1]**

(b) Substituting $x = 5$ into $(x + 2) + 2 + x + 3$ **[M1]**

17 **[A1]**

(c) Identifying the correct regions: $(2x) + 3$ **[M1]**

13 **[A1]**

(d) The region strictly inside F, but outside both T and S shaded. (The 'Football only' region) **[B1]**

[Back to Question] *[Total Marks: 8]*

14 (a) Intersection of all three: 8 AND Outer region: 3 **[B1]**

Two-set only intersections: 12, 7, and 10 in correct regions **[B1]**

Single set regions: x (in C), $2x$ (in D), and x (in B) **[B1]**

(b) $x + 2x + x + 8 + 12 + 7 + 10 + 3 = 100$ oe **[M1]**

$x = 15$ **[A1]**

(c) Identifies the regions required: Only C + Only D + ($C \cap D$ only) OR uses $x + 2x + 12$ **[M1]**

57 **[A1]**

[Back to Question] *[Total Marks: 7]*

15 (a) 4 in center intersection **[B1]**

8, 5, 7 in the two-sport intersections **[B1]**

13, 9, 9 in the one-sport regions AND 5 in outer region **[B1]**

(b) 5 **[B1]**

(c) 17 **[B1]**

(d) $B \cap T' \cap S'$ or $B \cap (T \cup S)'$ **[B2]**

M1 for $B \cap \ldots$ **[M1]**

[Back to Question] *[Total Marks: 7]*

16 (a) $2x$ in the T only region **[B1]**

x in the intersection and $3x$ in the C only region **[B1]**

2 in the region outside both circles **[B1]**

(b) 8 **[B2]**

M1 for $2x + x + 3x + 2 = 50$ or $6x = 48$.

(c) 16 **[B1]**

(d) Region outside T shaded, as well as the entirety of C (the only unshaded region is 'T only') **[B2]**

B1 for shading T' correctly OR shading C correctly.

[Back to Question] *[Total Marks: 8]*

17 (a) Places 15 in the central intersection **[B1]**

Places 15 in $B \cap T$ only, 13 in $T \cap C$ only, and 7 in $B \cap C$ only **[B1]**

Places x, $2x$, $x+5$, and 8 in the respective remaining regions (B only, T only, C only, and outside the sets) **[B1]**

(b) $x + 2x + (x+5) + 15 + 15 + 13 + 7 + 8 = 103 \implies 4x + 63 = 103$ (or equivalent valid method) **[M1]**

$x = 10$ **[A1]**

(c) 85 **[B1]**

(d) $\frac{13}{63}$ **[B2]**

If 0 scored, allow [M1] for correctly identifying the denominator $n(T) = 63$ or the numerator $n(T \cap C \cap B') = 13$

[Back to Question] *[Total Marks: 8]*

18 **(a)** All 8 regions correctly populated **[B3]**

Partial working: 5 to 7 regions correctly populated **[B2]**

Partial working: 3 to 4 regions correctly populated (e.g. places $12, 15, x$ correctly) **[B1]**

(Correct values are: Center = 12. $M \cap D$ only = 13. $D \cap A$ only = 20. $M \cap A$ only = 16. M only = 9. D only = 20. A only = x. Outside = 15)

(b) 15 **[B2]**

Partial working: Forms correct sum $9 + 20 + x + 13 + 16 + 20 + 12 + 15 = 120$ OR $105 + x = 120$ **[M1]**

(c) 70 **[B2]**

Partial working: Identifies the 20 students in only Drama and Art OR correctly calculates $50 + 20$ **[M1]**

(d) Region strictly inside M but outside both D and A is shaded (the top crescent / "only M" region) **[B1]**

[Back to Question] *[Total Marks: 8]*